Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

Mathematics I

BCA — Semester I — 2022 — Answer Sheet
Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. In a certain village in Nepal, all the people speak Nepali or Tharu or both languages. If 90% speak Nepali and 20% speak Tharu, how many people speak:i) Nepali language onlyii) Tharu language onlyiii) both languages

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. If $x - ty = \frac{5 - 6i}{5 + 6i}$, prove that $x^2 + y^2 = 1$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Define a function. Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3x + 5$ is bijective.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. If A is the A.M. and H is the H.M. between two numbers a and b, show that: $$\frac{a - A}{a - H} \times \frac{b - A}{b - H} = \frac{A}{H}$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define matrix. If $A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} -2 & 1 \\ 3 & 2 \end{pmatrix}$, show that: $(AB)^T = B^T A^T$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Prove that: $$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a - b)(b - c)(c - a)$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Find the eccentricity and foci of the ellipse: $25x^2 + 4y^2 = 100$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. a) A bag contains 8 red balls and 5 blue balls. In how many ways can 3 red balls and 4 blue balls be drawn?b) Find the volume of the parallelepiped whose concurrent edges are represented by the vectors $\vec{i} - 2\vec{j} + 3\vec{k}$, $-3\vec{i} + 4\vec{j} - 5\vec{k}$, and $\vec{i} + 2\vec{j} - 3\vec{k}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. a) Find the Taylor Series expansion of $f(x) = x^3 - 2x + 4$ at $a = 2$.b) In how many ways can the letters of the word 'CALCULUS' be arranged so that the two C's do not come together?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define exponential and logarithmic functions. If $f(x) = \log \frac{1 - x}{1 + x}$ for $-1 < x < 1$, show that: $$f\left(\frac{2ab}{1 + a^2 b^2}\right) = 2f(ab) \quad \text{where } |ab| < 1.$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
